Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 7 · day per timetable	7	1 period (60 min)	Exponential and Logarithmic Equations and Inequalities	Algebra II	Topic 6 — Exponential and Logarithmic Functions · Lesson 6-6	Grade 11 — 11B

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · GeoGebra Graphing Calculator · Calculator · Canva (final product) · Whiteboards	Lesson presentation (14 slides) · “Choose Your Route” differentiation sheet · “Give it a go” classwork · “Red Sea Resort Planning” project task (A3) · squared paper

Competencies	Technology Embedded	Connection to student's real life
1- Solving exponential equations and inequalities by matching bases or by using logarithms. 2- Solving logarithmic equations and inequalities and rejecting extraneous solutions by checking the domain.	1- Quizizz live diagnostic with instant gap analysis. 2- GeoGebra Graphing Calculator: graph both sides of an equation and read the intersection to check an algebraic answer.	1- Two Red Sea resorts growing at different rates: when does each reach its target? 2- Water use per visitor that falls each year: when does it drop below a target?

Values	Soft Skills	Hard Skills
1- Precision — checking every answer against the domain. 2- Sustainability — planning resources for national targets.	1- Collaboration in assigned group roles. 2- Explaining each step as following from the line before it.	1- Rewriting with a common base and solving. 2- Solving logarithmic equations and inequalities and using logarithms in models.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Solve exponential equations and inequalities by applying the concept of a common base. 2- Solve logarithmic equations and inequalities by using the rules of logarithms. 3- Use logarithms to solve exponential equations and models. *Essential question: How can we solve exponential and logarithmic equations and inequalities?* *Vocabulary: exponential equation ; exponential inequality ; logarithmic equation ; logarithmic inequality.*	FIKR Phase 1 — Diagnose. 3-question Quizizz on common bases, power of a power and evaluating logarithms. Results read as a gap map; students routed to re-teach, the examples, or Investigate. FIKR Phase 2 — Targeted Instruction. Re-teach common bases for flagged students. Whole class: exponential equations and inequalities (base-below-1 flip); logarithmic equations and inequalities (domain check); logarithms for models. FIKR Phase 3 — Practice. Two "Give it a go" problems worked together; students then choose a route and may switch at any time. Pear Deck feedback; 2-minute micro-conferences. FIKR Phase 4 — Production. The Red Sea resort model: a logarithm for one resort, matching bases for the other, an inequality in whole years. AI as critic only. FIKR Phase 5 — Proof of Learning. "Time to Check": a common-base equation, a logarithmic equation with a domain check, and a base-below-1 inequality. No notes, partner or AI. FIKR Phase 6 — Smart Production. Students refine their reasoning into a final product with one worked non-example, and publish it.	Diagnostic quiz — gap map, not a grade. Quick check for understanding on whiteboards (80% threshold). "Give it a go" questions — self-marked, teacher circulates. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. PASS → Enrichment & Challenge. NOT YET → Targeted Learning Clinic next lesson. Final product graded on reasoning and clarity, not correctness alone.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Peer show-and-tell (2 min), then the reflection: "Which is harder — matching the bases or choosing a logarithm?"
Reflection for Improvement:	
Homework:	Upload the final product to the LMS portfolio; Pear Deck practice set; flipped video for Lesson 6-7.
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